

An Ontology-Driven Graph RAG for Legal Norms: A Hierarchical, Temporal, and Deterministic Approach

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Abstract. Standard Retrieval-Augmented Generation (RAG) systems fail in the legal domain, as they are blind to the hierarchical, diachronic, and causal structure of law, leading to anachronistic and unreliable answers. This paper introduces the Structure-Aware Temporal Graph RAG (**SAT-Graph RAG**), an ontology-driven framework designed to overcome these limitations by explicitly modeling the formal structure and diachronic nature of legal norms. We ground our knowledge graph in a formal, LRMoo-inspired model that distinguishes abstract legal Works from their versioned Expressions. To validate this architecture, we present a qualitative, trace-based evaluation demonstrating how an agentic framework can leverage our model's deterministic primitives to resolve complex temporal and provenance queries that are intractable for standard RAG. The result is a practical and auditable framework for building trustworthy and explainable legal AI systems.

Keywords. Graph RAG, Legal Knowledge Graphs, Temporal Modeling, Provenance Reconstruction, Computational Law, Deterministic Retrieval, LRMoo

1. Introduction

Retrieval-Augmented Generation (RAG) systems [1], while promising, fail to address the unique challenges of legal texts. Legal norms are not flat documents; they possess a formal hierarchy, a dense web of cross-references, and a diachronic evolution through amendments and repeals. This interplay between structural hierarchy and temporal dynamism is a primary obstacle for standard AI, as a “structurally and temporally-naïve” system cannot deterministically retrieve the version of any legal provision valid on a specific date, leading to factually incorrect answers unacceptable in a high-stakes domain.

To address this, we introduce the Structure-Aware Temporal Graph RAG (SAT-Graph RAG), an ontology-driven framework grounded in LRMoo [2]. It models the legal norm's multi-layered structure, propagates changes efficiently via aggregation, and reifies legislative events as queryable Action nodes, enabling a multi-aspect retrieval where a norm's causal history and metadata become as searchable as its textual content. This paper presents this conceptual blueprint and a proof-of-concept evaluation, demonstrat-

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ing how the architecture—formalized in the SAT-Graph API [3]—enables deterministic answers to complex temporal queries that are intractable for standard RAG.

We review related work in Section 2, detail our framework in Section 3, present a proof-of-concept evaluation² in Section 4 and conclude in Section 5.

2. Related Work

Research in RAG [1] and Graph RAG [4] confirms that external knowledge improves performance. Legal benchmarks like LegalBench-RAG highlight retrieval as a key bottleneck [5], as naive approaches ignore the structural information from legislation that is crucial for statutory retrieval and QA [6,7]. While research on Temporal Knowledge Graphs (TKGs) has focused on representation and reasoning [8], few works integrate the hierarchical and cross-referential structure of legal texts with robust mechanisms for provenance and historical versioning [9]. Legislative standards like Akoma-Ntoso [10] aim to capture these complexities, confirming that legal representation demands explicit temporal and hierarchical support [11]. However, a systematic integration of hierarchical versioning with temporally-aware, provenance-based retrieval remains a critical gap. Our work addresses this gap by grounding a Graph RAG architecture in a formal ontology, enabling deterministic, temporally-aware retrieval with auditable provenance, inspired by hybrid approaches that fuse structured and textual data to improve explainability [12].

3. The Proposed Framework

3.1. Ontological Foundation and Graph Construction

Our framework models the hierarchical, temporal, and linguistic complexities of legal norms using a graph structure grounded in a formal model inspired by the IFLA LRMoo ontology [2], which elevates the Work/Expression distinction to first-class entities. While traditional FRBR, designed for bibliographic records, can be interpreted more restrictively, LRMoo is explicitly designed for interoperability and permits modeling temporal and linguistic variants as distinct Expressions. Following legislative informatics practice, we model each dated version of a norm as a distinct Expression. The full diachronic modeling principles are detailed in a foundational work [13], currently available as a preprint; here, we define four primary node types:

Norm: An abstract legal creation (e.g., “1988 Brazilian Constitution”). (LRMoo: Work)

Component: A hierarchical element within a norm (e.g., a title, chapter, or article). (LRMoo: Work)

Temporal Version (TV/CTV): A language-agnostic “temporal snapshot” of a Norm or Component, capturing its semantic content. (LRMoo: Expression)

Language Version (LV/CLV): The concrete textual realization of a TV/CTV in a specific language. (LRMoo: Expression)

²A formal data model and API contract is provided as an OpenAPI 3.0 specification, publicly available at: <https://github.com/hmartim/sat-graph-api>

Graph construction begins with structure-aware semantic segmentation. This process identifies the abstract structural entities, creating the backbone of **Norm** and **Component** nodes (Figure 1). However, linking the textual content (the *Text Unit*) directly to a timeless Component node would be flawed, as it fails to capture the law’s diachronic nature: the abstract Component is permanent, but its textual expression is not.

Our model resolves this by using versioning layers. For the initial enactment and for each subsequent amendment, we instantiate a Component **Temporal Version** (CTV) representing its semantic content on a specific date, and a corresponding Component **Language Version** (CLV) for its concrete wording. The text itself is exclusively associated with the CLV, ensuring every piece of content is unambiguously tied to both a specific semantic state and a linguistic expression (Figure 1). This design efficiently scales to multilingual corpora by reusing the abstract and temporal nodes, only creating new language-specific CLVs and their associated **Text Units** for each translation.

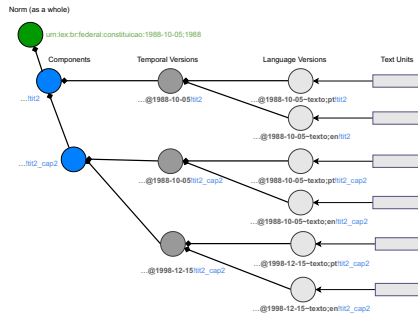


Figure 1. The model’s layers: Norms (green), Components (blue), Temporal Versions (dark gray), Language Versions (light gray), and Text Units (rectangles).

3.2. Temporal Versions as Aggregations

A key innovation is how our framework propagates changes by replacing naive **Composition** with efficient **Aggregation**. A compositional model, which creates new CTVs for *all* children of an amended component (even those unchanged), is highly inefficient, creates redundant data, and obscures the actual modifications. Our aggregation model for a parent CTV on date D_n simply reuses the *most recent available* CTVs of its unchanged children. This economical approach enables the reuse of child CTVs across parent versions and guarantees the deterministic reconstruction of any provision on a specific date.

3.3. Modeling Causality and Metadata as Retrievable Units

To make the graph’s semantic capabilities searchable, we model normative events and metadata as first-class entities. To reify the causality of legislative events, we introduce an **Action** node for each granular change, grounded in event-centric legal ontologies [13, 14]. Each Action represents a specific command from a normative instrument (e.g., an amendment) and acts as a formal causal link, connecting the source provision’s original CTV to its effect on the target norm: terminating one CTV/TV and creating another.

We generate a separate, descriptive *Text Unit* for the *Action* itself. This structured summary textualizes the event’s key details—such as the amending law, effective dates, and a description of the change. This transforms the legislative event into a searchable object, enabling queries about a law’s history and causal evolution, not just its content.

We apply the same principle to structured metadata. Information like publication dates or succession relationships, normally invisible to text-based retrieval, is converted into natural language **Metadata Text Units**, linked to Norm, or Component or Temporal Version, nodes. This operationalizes multi-aspect representation [15], making a norm’s entire context—from causal events to metadata—as retrievable as its content.

3.4. Structure-Aware Retrieval via Curated Communities

Departing from standard Graph RAG approaches that rely on algorithmic community detection, our framework leverages two forms of curated communities intrinsic to the legal domain. The first is the norm’s own **Internal Hierarchy**: its predefined structure (Titles, Chapters) that provides a natural, nested hierarchy where each grouping component serves as a community. The second is the **External Thematic Classification**: human-curated legal taxonomies (e.g., “Social Security”) which are modeled as high-level **Theme** nodes and group related norms and components across the corpus. These nodes can form their own hierarchy, allowing broader themes.

This curated structure enables a powerful filtering mechanism: a user can select any node—a Theme, Norm, Component, or Temporal Version—to define a precise query scope. The semantic search is then performed exclusively on the contextually relevant subset of *Text Units* associated with that scope. This transforms retrieval from a flat search into a process of semantic navigation, yielding more efficient and precise results.

4. Proof-of-Concept Evaluation

We validate our framework with a proof-of-concept evaluation using the Brazilian Federal Constitution of 1988, a text with a complex diachronic history. While our conceptual model (Section 3) distinguishes four layers for clarity, this evaluation uses the pragmatic, streamlined model from our SAT-Graph API [3], where Norm and Component are unified into **Item**, the Temporal Version (TV/CTV) is simplified to **Versión** and all associated text is collapsed into **TextUnit**. This simplification reduces engineering complexity without compromising the deterministic capabilities demonstrated below.

We present a qualitative, trace-based simulation of an agentic framework resolving complex queries, contrasting its outputs with a baseline representing the dominant paradigm in current RAG deployments: a flat index of a document’s current version. Its predictable failures are not presented as a straw man, but serve to illustrate the fundamental architectural limitations our causally-aware framework is designed to overcome.

4.1. Modeling a Legislative Amendment: The Case of Article 6

Our evaluation focuses on Article 6 (more precisely, on its caput), which enumerates social rights and has been amended multiple times. For instance, Constitutional Amendment (CA) No. 26/2000-02-14 added the right to “housing” (*moradia*). In our graph, this is modeled as a discrete Action node that: (1) is causally linked to the Version of the CA

26 provision Item that commanded the change; (2) terminates the original 1988-10-05 Version of Article 6's caput; and (3) creates a new 2000-02-14 Version for Art. 6's caput, linked to a new TextUnit containing the updated pt-BR wording.

4.2. Answering Complex Temporal Queries

Query Pattern 1: Point-in-Time Retrieval

For a query like “*What were the social rights listed in Article 6 of the Brazilian Constitution on May 20th, 1999?*”, a standard RAG fails by retrieving the current text, which includes rights added after that date. Our framework succeeds through a deterministic, three-step agent plan: (1) resolve “Article 6, caput of the Brazilian Constitution” to its canonical Item ID (e.g., via the `resolveItemReference` API call); (2) find the single Version valid on the target date (1999-05-20) (e.g., via `getValidVersion`); and (3) retrieve its exact TextUnit (e.g., via `getTextForVersion`). The LLM thus receives only the temporally-correct text, enabling a factually accurate, auditable answer.

Query Pattern 2: Provenance and Causal-Lineage Reconstruction

To trace the legislative lineage of a term like “food” in Article 6, a baseline RAG's semantic search is probabilistic and cannot guarantee retrieval of the causal law, nor can it provide a verifiable “before/after” comparison. Our framework enables a deterministic agent plan: (1) resolve “Article 6, caput of the Brazilian Constitution” to its Item ID (e.g., via `resolveItemReference`); (2) retrieve its full history of Action nodes (e.g., via `getItemHistory`); (3) identify the pivotal Action that introduced the term by inspecting the text of each resulting Version (e.g., via `getBatchTexts`). This process yields structured, verifiable data (the amending law, date, and textual difference) for an auditable report.

5. Discussion and Conclusion

This paper introduced an ontology-driven Graph RAG architecture that transforms legal retrieval from a probabilistic search into a deterministic, auditable process. Grounded in a formal, LRMoo-inspired model, our principal contributions—explicitly modeling legal structure, versioning, and causal Actions—provide a retrieval substrate superior to flat RAG systems for trustworthy legal AI [16]. While this determinism requires significant data curation, our aggregation strategy demonstrates promising scalability; for a corpus like the Brazilian Constitution, the graph's growth is sublinear relative to the number of amendments—our aggregation strategy for Constitution produced a graph with 5121 Versions (TV/CTVs) across 3085 Items (Norm/Components) modified by 136 Constitutional Amendments. This approach lays a necessary foundation for high-stakes legal AI, though its output quality depends on accurate upstream data.

While this paper's qualitative proof-of-concept validates our deterministic logic, future work will focus on large-scale quantitative evaluation against legal AI benchmarks and the publication of a formal OWL/RDF ontology to support semantic interoperability.

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